PYTHON LAB ASSIGNMENT EXPERIMENT-1

BY

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BATCH:- 1

1. Install Python and write the steps of installation and understand difference between scripting and interactive modes in IDLE.

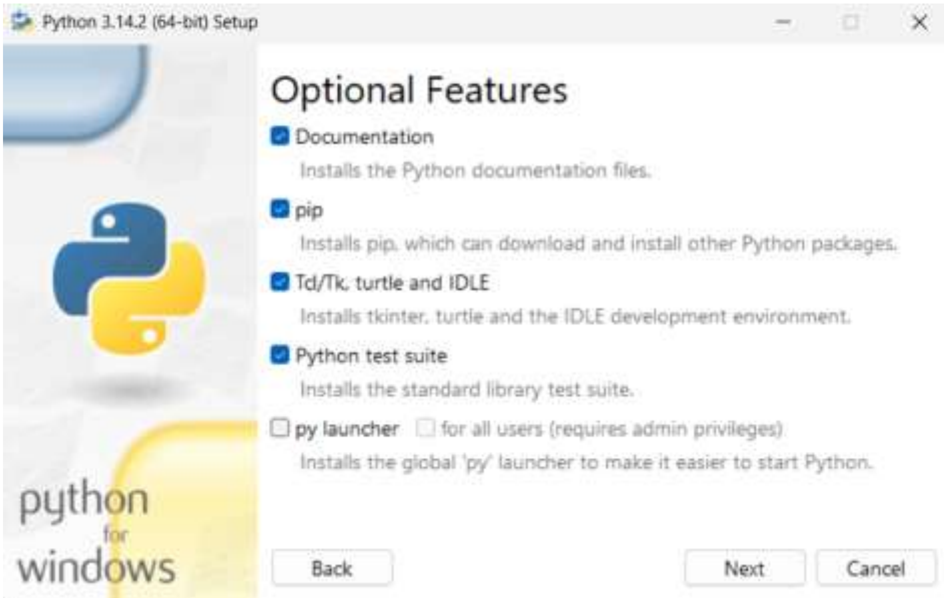
-> Python Installation:

First, download Python from www.python.org. Open the setup file, check “Add Python to PATH”, then click Install Now. After installation, open Command Prompt, type python, and if the version appears, Python is installed successfully.

Difference between Interactive and Scripting Mode in IDLE:

In Interactive Mode, Python runs one line at a time and gives output immediately, which is useful for practice and testing.

In Scripting Mode, we write the complete program, save it as a .py file, and run it together, which is used for writing real programs.



2. Create a variable to store your age and print its type using type().

➔ Code:-

```python
#Raghav Vij 590023842
age = 20
print(type(age))
```

➔ Output:-

```
PS C:\Users\hp> p
<class 'int'>
PS C:\Users\hp>
```

3. Declare a string variable called x and assign it the value “Hello”.

a) Print out the value of x

➔ Code:-

```python
q3.py > ...
#Raghav Vij 590023842
x = "Hello"
print(x)
```

➔ Output:-

```
PS C:\Users\hp>
Hello
PS C:\Users\hp>
```

4. Take different data types and print values using print function.

➔ Code:-

```python
#Raghav Vij 590023842
a = 10
b = 3.14
c = "Hello"
d = True

print(a)
print(b)
print(c)
print(d)
```

➔ Output:-

```
10
3.14
Hello
True
PS C:\Users\hp>
```

**5.** Declare these variables (x, y and z) as integers. Assign a value of 9 to x, Assign a value of 7 to y, perform addition, multiplication, division and subtraction on these two variables and Print out the result.

➔ **Code:-**

```
#Raghav Vij 590023842
x = 9
y = 7

print("Addition:", x + y)
print("Multiplication:", x * y)
print("Division:", x / y)
print("Subtraction:", x - y)
```

➔ **Output:-**

```
Addition: 16
Multiplication: 63
Division: 1.2857142857142858
Subtraction: 2
PS C:\Users\hp>
```

6. Write a program to compute the length of the hypotenuse (c) of a right triangle using Pythagoras theorem.

➔ Code:-

```
q6.py > ...
#Raghav Vij 590023842
a = float(input("Enter base: "))
b = float(input("Enter height: "))

c = (a**2 + b**2) ** 0.5

print("Length of hypotenuse:", c)
```

➔ Output:-

```
Enter base: 10
Enter height: 10
Length of hypotenuse: 14.142135623730951
PS C:\Users\hp\Documents\UPES\SEM-2\PYTH
```

7. Write a program to find simple interest.

➔ Code:-

```python
#Raghav Vij 590023842
p = float(input("Enter principal amount: "))
r = float(input("Enter rate of interest: "))
t = float(input("Enter time (in years): "))

si = (p * r * t) / 100

print("Simple Interest:", si)
```

➔ Output:-

```
Enter principal amount: 10000
Enter rate of interest: 10
Enter time (in years): 12
Simple Interest: 12000.0
```

8. Write a program to find area of triangle when length of sides are given.

➔ Code:-

```python
q8.py > ...
#Raghav Vij 590023842
a = float(input("Enter first side: "))
b = float(input("Enter second side: "))
c = float(input("Enter third side: "))

s = (a + b + c) / 2
area = (s * (s - a) * (s - b) * (s - c)) ** 0.5

print("Area of triangle:", area)
```

➔ Output:-

```
Enter first side: 10
Enter second side: 10
Enter third side: 10
Area of triangle: 43.30127018922193
```

➔

9. Write a program to convert given seconds into hours, minutes and remaining seconds.

➔ Code:-

```python
#Raghav Vij 590023842
sec = int(input("Enter seconds: "))

hours = sec // 3600
minutes = (sec % 3600) // 60
seconds = sec % 60

print("Hours:", hours)
print("Minutes:", minutes)
print("Seconds:", seconds)
```

➔ Output:-

```
Enter seconds: 3600
Hours: 1
Minutes: 0
Seconds: 0
```

➔

10. Write a program to swap two numbers without taking additional variable.

➔ Code:-

```python
#Raghav Vij 590023842
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

a, b = b, a

print("a =", a)
print("b =", b)
```

➔

➔ Output:-

```
Enter first number: 10
Enter second number: 20
a = 20
b = 10
```

➔

11. Write a program to find sum of first n natural numbers.

➔ Code:-

```python
#Raghav Vij 590023842
n = int(input("Enter n: "))

sum = n * (n + 1) // 2

print("Sum:", sum)
```

➔

➔ Output:-

```
11.py"
Enter n: 5
Sum: 15
```

➔

12. Write a program to print truth table for bitwise operators (&, | and ^ operators)

➔ Code:-

```python
#Raghav Vij 590023842
a = 1
b = 0

print("a b  a&b  a|b  a^b")
print(a, b, a & b, a | b, a ^ b)

a = 0
b = 1
print(a, b, a & b, a | b, a ^ b)

a = 1
b = 1
print(a, b, a & b, a | b, a ^ b)

a = 0
b = 0
print(a, b, a & b, a | b, a ^ b)
```

➔

➔ Output:-

```
12.py"
a b  a&b  a|b  a^b
1 0 0 1 1
0 1 0 1 1
1 1 1 1 0
0 0 0 0 0
PS C:\Users\hp\Documents
```

13. Write a program to find left shift and right shift values of a given number.

- Code:-

```python
#Raghav Vij 590023842
n = int(input("Enter number: "))

print("Left shift:", n << 1)
print("Right shift:", n >> 1)
```

- Output:-

```
13.py"
Enter number: 1
Left shift: 2
Right shift: 0
```

14. Using membership operator find whether a given number is in sequence (10,20,56,78,89)

- Code:-

```python
#Raghav Vij 590023842
n = int(input("Enter number: "))

seq = (10, 20, 56, 78, 89)

print(n in seq)
```

- Output:-

```
14.py"
Enter number: 10
True
```